PROPOSTA DI GRANT GIOVANI IN CSN5 ANNO 2024

Nome: OPTIMAL
Titolo: automated Optimization of x-ray beamline Parameters using experimenT-Informed MAchine Learning methods
Proponente: Simone Manti
Unità partecipanti: INFN Laboratori Nazionali di Frascati

Abstract

X-ray beamlines serve a variety of purposes, ranging from fundamental physics research to practical societal applications, each requiring different energetic and spatial resolutions, signal intensity and measurement's time scale. Optimizing beamline performances to meet these needs, and addressing their complex multi-parameter space components is usually time-consuming and costly. In this context, the OPTIMAL project aims to deliver a framework for automatic optimization of X-ray beamlines by pioneering the use of experiment-informed Bayesian Optimization (BO) and Reinforcement Learning (RL). As a test bench, an experimental setup with 10 parameters for X-ray Emission Spectroscopy (XES), employing a Bragg Von Hamos spectrometer with a two slits configuration, will be used, with an expected 10-fold improvement in acquisition time. For this setup, the achievable energy resolution, stability, and intensity—strongly depend on the source, crystal and position detector, making it an ideal platform for developing the OPTIMAL framework. Specifically, BO is suitable for scenarios with costly objective functions, and uses Gaussian Processes (GP) to model uncertainties and update predictions based on the recorded data. Alternatively, RL will be used to continuously adapt parameters in real-time, using feedback from XES spectra to dynamically improve the configuration. To further enhance the performance, Machine Learning, in the form of Deep Learning, will extract features from XES spectra and integrate them into the BO optimization loop or in the RL algorithm. OPTIMAL will deliver an optimized XES spectrometer ready for practical use and capable of balancing trade-offs between acquisition time, energy precision, and resolution. OPTIMAL will establish a versatile optimization framework applicable to a wide range of X-ray measurement applications and fully extendable to X-ray beamlines and facilities such as DAΦNE-Light, EuAPS, and, in future, to EUPRAXIA.

Contents

1 State of the art

X-ray beamlines are advanced tools that heavily rely on the optimization of interconnected parameters, to ensure the beam quality and the accuracy of the results. For instance, one needs to optimize the alignment of optical components to achieve the best energy and spatial resolutions and minimize the acquisition time. Consequently, optimization tasks are integral and crucial to every phase of a beamline's experiment lifecycle, from its design, during the measurement, up to data analysis.
A naive grid search of the optimal parameters becomes prohibitively time and cost consuming as the dimensionality of the parameter space increases, and a random search does not always guarantee to find the desired extremum or provide control over the optimization process. For example, in beamline experiments where time is limited, it is crucial to focus on performing the measurements and minimize the time spent on optimization to make the most efficient use of the available time. **Thus, optimization methods for finding the optimal settings of experiments are of great importance**.
Advancements in Artificial Intelligence (AI) and increased computational resources have revolutionized these fields, and Machine Learning (ML) algorithms can now directly learn from data to help identifying the optimal settings.
For instance, Bayesian Optimization (BO) methods [1–3] utilize surrogate models, typically Gaussian Processes (GP), to learn from data and guide the optimization loop towards the ideal configuration. A more advanced approach is Multi-Objective Bayesian Optimization (MOBO), which simultaneously considers multiple variables by identifying the Pareto front of the optimization. Additionally, incorporating ML techniques, such as using Deep Neural Networks as a filter for the GP for features extraction, **can enhance the optimization process**, especially for structured data.
For this reason, BO has proved its success in various domains [4]. Restricting to the field of physics, BO has been extremely useful for particle accelerators [5, 6], both in setting up parameters, as well as during the data-taking process. It also helps in the design of particle detectors [7] that work in connection with realistic simulation of the detector. BO applications are as well important in beamlines, for example in optimizing laser pulse in Free-Electron Laser [8] or tuning the beam's dimension for generating plasma [9] in plasma accelerator.
For X-ray beamlines, **BO has seen fewer applications compared to other areas**, but its usage is gradually increasing [10]. For instance, BO has been successfully applied in X-ray Absorption Spectroscopy (XAS) to optimize beamline parameters, which is essential for maximizing the detection efficiency and minimizing the acquisition time [11]. Furthermore, precise alignment of components, including sample positioning and environmental conditions, is essential for achieving high energy resolution and accuracy [12, 13].
Bayesian Optimization (BO) can become challenging when the number of parameters is extremely high. In contrast, Reinforcement Learning (RL) offers a fast alternative by building a black-box model of the experiment and guiding the optimization process through iterative feedback. This approach often requires a numerical simulation of the experiment to accurately model the system dynamics and provide the necessary feedback for the RL algorithm. For example, an RL agent has been used to determine the optimal configuration of the ADIGE Medium Resolution Mass Separator at INFN Legnaro National Laboratories [14].
Given the advancements in optimization techniques and the identified gaps and limitations in the current state-of-the-art for the X-ray beamlines case, **it is the right time to develop ML-aided techniques to enhance their efficiency and precision, paving the way for more accurate and cost-effective experiments.**

2 Objectives (O) and Expected Results (ER)

O1: Within OPTIMAL, I will develop and implement a comprehensive framework for optimizing experiments performed on X-ray sources. This framework will include modules for data acquisition, setup control, simulations, and multi-objective optimization. I will utilize two techniques, BO and RL, both employing ML models trained on ray-tracing simulations of the experiments.

ER1: This will result in a versatile software platform for X-ray experiments, capable of performing optimizations, based on the beamline parameters and on the accuracy of the available simulations for each specific case.

O2: I will set up an experimental setup for X-ray Emission Spectroscopy (XES) with 10 adjustable parameters [see Figure (4)], that will be capable of achieving an energy precision below 0.1 eV and an energy resolution in the 1-10 eV range.

ER2: This setup will represent a real test bench to develop and test the OPTIMAL framework.

O3: I will optimize the 10 parameters of the XES setup, to achieve the desired energy precision and resolution of the XES spectra, with an expected 10-fold reduction in the acquisition time.

ER3: This optimization will result in an enhanced version of the setup compared to normal manual operations, characterized by a reduced human-user input, and it will serve as a testing system for different optimization methods.

O4: By the end of the project, I will assess the potential applications of the optimized XES setup in specific cases and evaluate the feasibility of integrating the OPTIMAL framework as an example at the DAΦNE-Light beam facility at LNF, utilizing conventional X-ray sources.

ER4: The anticipated outcome is to identify and document potential applications or experiments, both within LNF and externally, that can benefit from these enhancements.

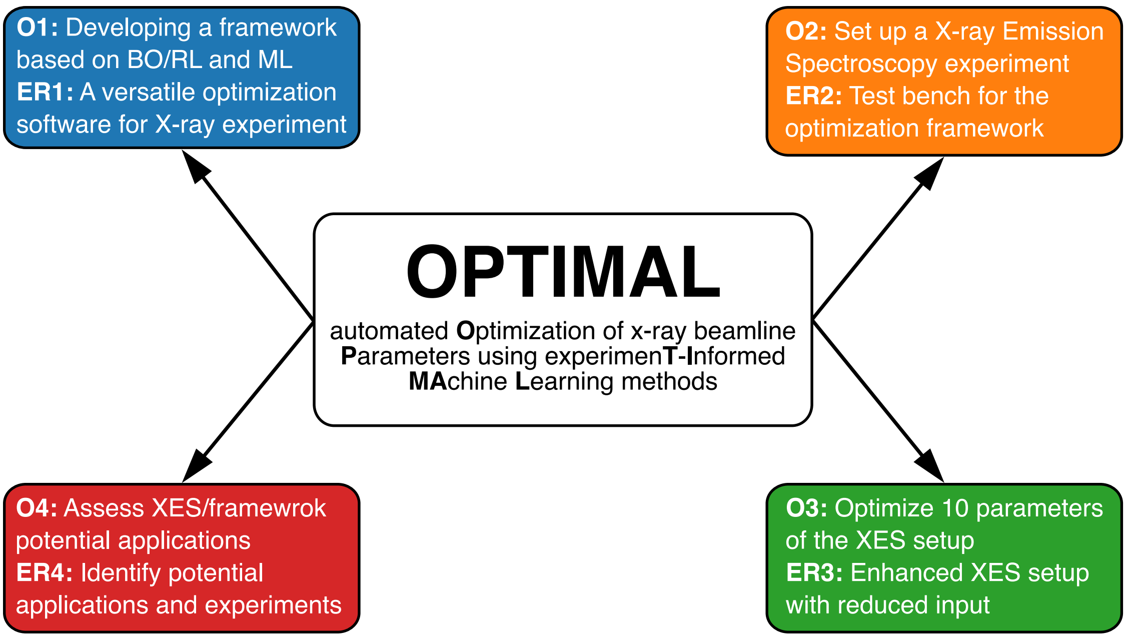

Figure 1: Objectives of the OPTIMAL project. The diagram illustrates the four objectives of the OPTIMAL project, starting from developing the framework and building a setup for XES, up to optimizing the spectra acquisition to propose application for X-ray beamlines.

3 Methodology

The goal of the OPTIMAL project is to develop a comprehensive framework for optimizing the parameters of X-ray line-based experiments, with a pilot application for a Bragg Von Hamos spectrometer. Below, I outline the methodological plan to achieve this objective, referencing the Work Packages (WP) defined in the next section.

Parameters optimization for XES

I will create the OPTIMAL framework (**WP1**) by applying and testing it for a XES experiment. Preliminary studies I performed at the LNF studying Fe$K\alpha$ emission line have shown that optimizing parameters for XES can lead to a significant improvement of the emission line's rate under study, with a gain factor of 10 in the overall acquisition time. More details are in the caption of Figure (2).

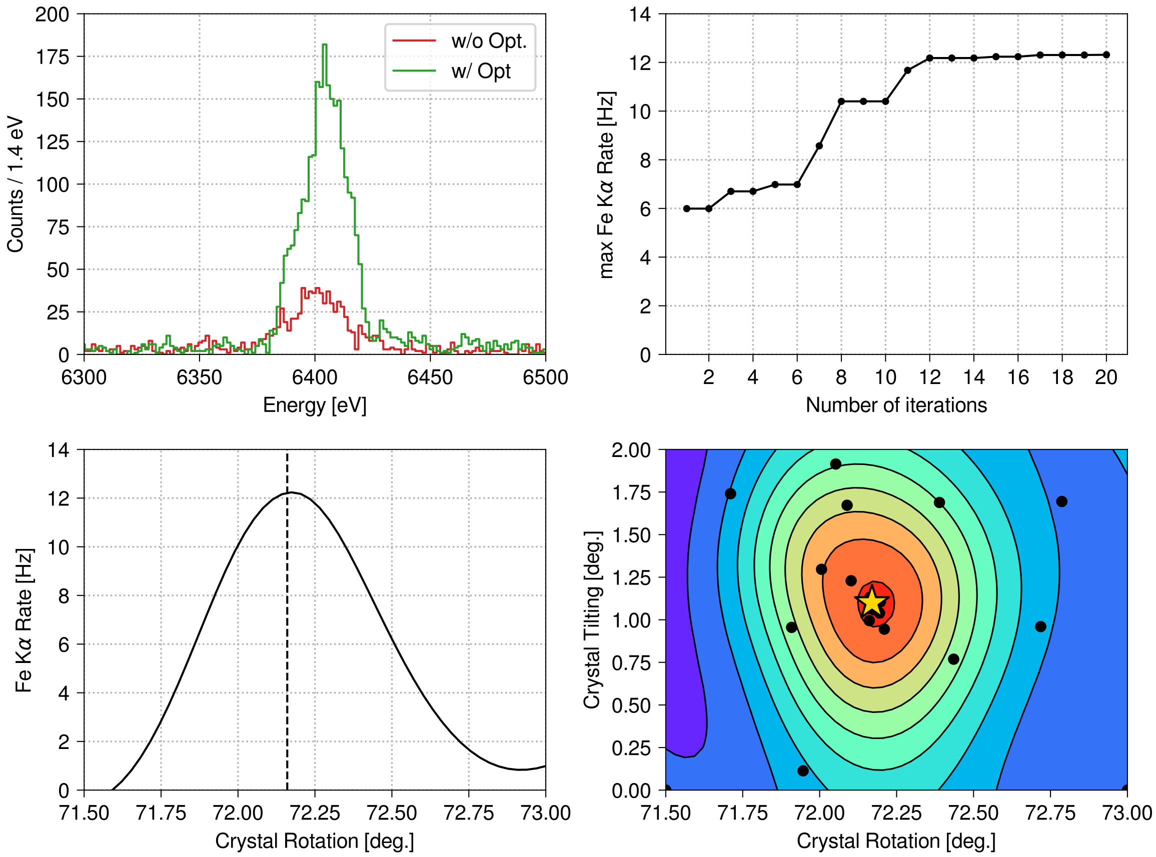

Figure 2: Bayesian optimization of iron's Kα. The plot on (*top left*) displays the Fe$K\alpha$ before and after the optimization of the rate, while on (*top right*) the maximum obtained rate is plotted as a function of the number of iterations, where for each iteration the rate is obtained from a fit of the spectrum. During the Bayesian optimization, the rotation and tilting of the crystal [see Figure (4)] are adjusted to maximize the rate. The associated Gaussian Process is plotted as a function of the rotation in the (*bottom left*) plot, while the partial dependence plot for the two parameters is displayed in the (*bottom left*), with the maximum indicated with a star. The optimization process aims to find the optimal crystal rotation and tilting angles to maximize the Kα peak rate of iron. With a precision of 0.01° within the specified angle ranges, the optimization provides an improvement of over a factor of 10 compared to a simple grid search.

I will develop further this idea in several ways. I will implement a Single-Objective (SO) optimization, focusing on three key quantities: rate signal, peak precision, and energy resolution of the spectra.

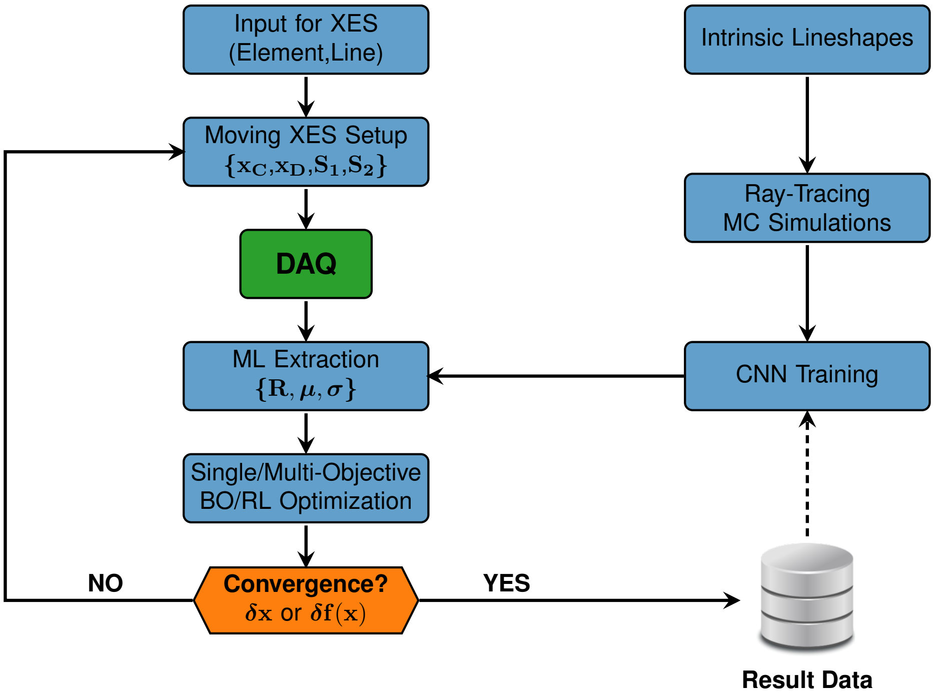

Figure 3: OPTIMAL's workflow. The OPTIMAL project workflow optimization involves several key steps. First, the framework adjusts the setup based on the selected input emission line and initiates the measurement with the DAQ, where $\mathbf{x_C}$ and $\mathbf{x_D}$ are collective coordinates for the crystal and detector [see Figure (4)]. Spectral feature extraction is done using ML models, previously trained on ray-tracing simulations, to extract the target properties for optimization. The optimization loop continues until convergence, depending on the threshold over the parameters $\mathbf{x}$ or on the objective functions $\mathbf{f}(\mathbf{x})$, then the results are saved in a database, that eventually can be used for training on historical real data.

Next, I will generalize to the Multi-Objective (MO) optimization case by considering all the peak's properties simultaneously. In this case, the goal is to find the Pareto front [3], which represents the set of optimal solutions where no objective can be improved without worsening at least another one. Finally, unlike the pilot case study, where the rate signal was manually extracted with a fit, I will generalize this objective extraction process with the aid of ML. The workflow I will implement in the OPTIMAL project is illustrated in Figure (3) with more details in the caption. In the following, I will describe the methodology of its components.

Experimental Setup for XES

To optimize the experimental setup for XES (**WP2**), **I will use several in-kind contributions from the National Laboratory of Frascati (LNF) of INFN**. Specifically, the infrastructure from the VOXES project will be utilized. The most significant contribution is a bunker, with Pb-PVC sandwich walls, located in building 24 of LNF, where experiments with X-rays can be safely conducted. Additional contributions will include essential elements of the XES setup, such as an X-ray tube, mosaic crystals, motorized stages, two slits, and a DECTRIS Mythen2 strip detector [see Figure (4)]. **The main financial request for this proposal concerns the upgrade of the VOXES spectrometer with a hybrid pixel detector, a crucial aspect of the project**. The optimization task, requires extracting features of interest from the acquired spectrum, like the iron peak in Figure (2). The current VOXES setup is inadequate for this purpose, as it utilizes a strip detector that only provides a one-dimensional (1D) histogram of the spectrum. This 1D data limits the effectiveness of ML feature extraction for the optimization loop, as it lacks sufficient information for the training process. In contrast, a pixel detector with a two-dimensional

spectrum offers significantly more information, enhancing the training phase, and thus the spectral feature extraction part. Additionally, to correctly perform XES for very high energy resolution, it is crucial to ensure that the emission lines on the detector are not tilted and a simple strip detector cannot provide this information along the vertical direction.
The most balanced solution between cost and project requirements is the AdvaPIX Timepix3 from ADVACAM (Prague, Czech Republic). This pixel detector operates within an energy range of 5.4 to 20 keV, with a sensor thickness of 300 μm, and a pixel size of 55 μm, making it well-suited for the project's requirements. The chosen detector will be controlled by a framework via dedicated software and integrated into the acquisition system using Python. The detector produces images with dimensions of 512 $\times$ 256 pixels. Based on this, I estimated that a GPU with 16 GB of memory will be sufficient for both training and inference. This conclusion is supported by the observation that XES spectra predominantly consist of simple-feature spots, corresponding to the peaks in the spectrum. Consequently, the ML architecture required to learn this relatively simple structure does not need to have a large number of deep layers.
Due to the increased weight of the new pixel detector, the holding motorized stages need to be changed. Therefore, a 4-axis motorized stage system is required. I choose those from STANDA company, like the old ones, for their easy communication interface with the framework using the `libximc` package. Here, my expertise and that of Alberto Clozza, an LNF researcher, will be helpful, as he has extensive knowledge in interfacing components for data-taking. Additionally, a design development with the production of mechanical parts will be conducted to connect the detector, motorized stages, and the existing setup. This part will be produced in collaboration with the mechanical department of LNF.
The existing setup and the new detector components will be integrated with the development of a Data Acquisition (DAQ) system. This integration will be crucial for precisely controlling experimental parameters during acquisition, such as the partial image spectrum and control conditions, for interfacing with the optimization procedure and saving results in a database. The project will also benefit from the specialized knowledge of Massimiliano Bazzi, an electronic engineer of LNF, who will manage the integration of the detector with the electronic components of the existing setup.
For the application of the framework to XES experiment (**WP3**), out of the ten setup parameters, at least seven will be considered for optimization, and their relative importance can be assessed using the fANOVA algorithm [15]. Here I already expect that the most critical parameter will be the rotation of the crystal R_C, as it determines whether X-rays impinge on the detector and fulfill Bragg's rule. Other significant parameters include the tilting of the crystal, which affects the y-coordinate of the detector, and the rotation of the detector R_D, which influences the arrival position of the photon on the pixel detector space of the peaks. Additionally, the x-coordinate of the detector and the slit apertures S_1 and S_2 are crucial, with the slits having the strongest impact on the energy resolution.

Optimization Framework

For optimization, **the project will employ two different, but also interoperable, methods: Bayesian Optimization (BO) and Reinforcement Learning (RL)**. BO was identified as the ideal method for optimizing X-ray beamline parameters for several reasons. Firstly, it is well-suited for problems where the acquisition function to be optimized is time-consuming and contains noise. This is precisely the case for X-ray beamline, as in the proposed XES application, which is performed with a monochromator crystal, resulting in an acquired spectrum with an overall rate on the order of Hz. By optimizing a target function, BO finds the set of parameters that minimize or maximize the function. To achieve this, BO uses a surrogate model constructed with a Gaussian Process (GP), which provides a cost-effective approximation of the function and automatically embeds uncertainties due to its probabilistic nature [see Figure (5)]. For developing the optimization framework (**WP1**), I will use the `botorch` [16] Python package, which will be evaluated for its capabilities in handling optimization tasks.
The disadvantage of BO is that the number of required evaluation points increases dramatically if the number of parameters becomes too large. Therefore, the OPTIMAL framework will implement RL as a complementary approach, particularly suitable for scenarios requiring a large number of control

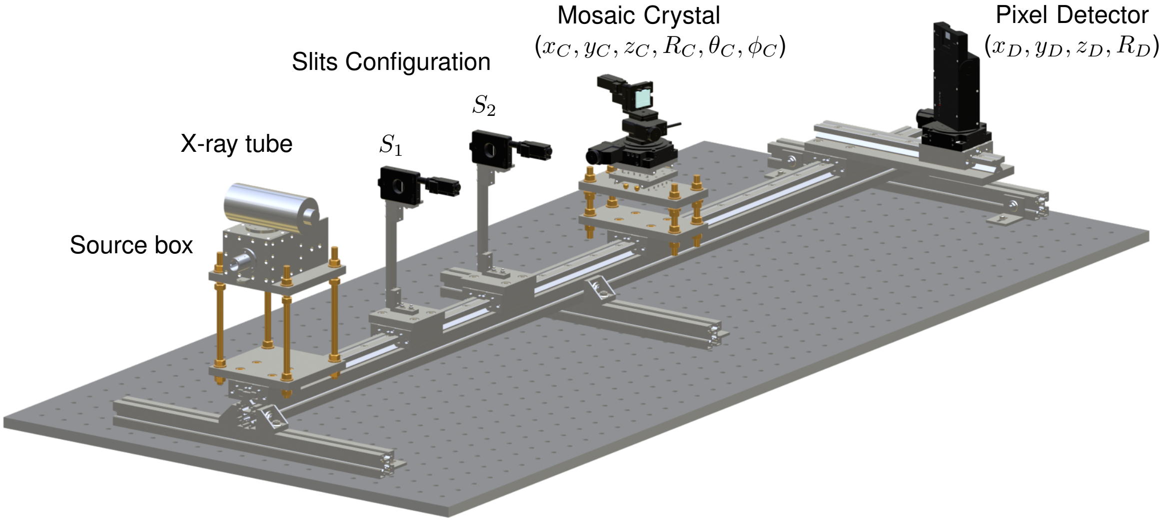

Figure 4: Proposed setup for XES. The proposed experimental setup for XES. The 10 parameters to be optimized in the project include the slit configuration, specifically the apertures of the two slits, S_1 and S_2. For the crystal, the parameters are the spatial coordinates (x_C, y_C, z_C), the rotation R_C around its z axis, and two tilting angles (θ_C, ϕ_C). For the pixel detector, the parameters are the spatial coordinates (x_D, y_D, z_D) and the rotation R_D around its z axis.

parameters. In such cases, RL builds an agent that interacts with the environment (in this case, the experiment) and takes action based on a policy designed for reward optimization. Here, the state is defined by the properties of the spectrum, and the policy aims to minimize the target property. RL's effectiveness relies on the accuracy of the simulations of the experiment, which **I will prove to be feasible for the proposed XES application**, even if not universally applicable.

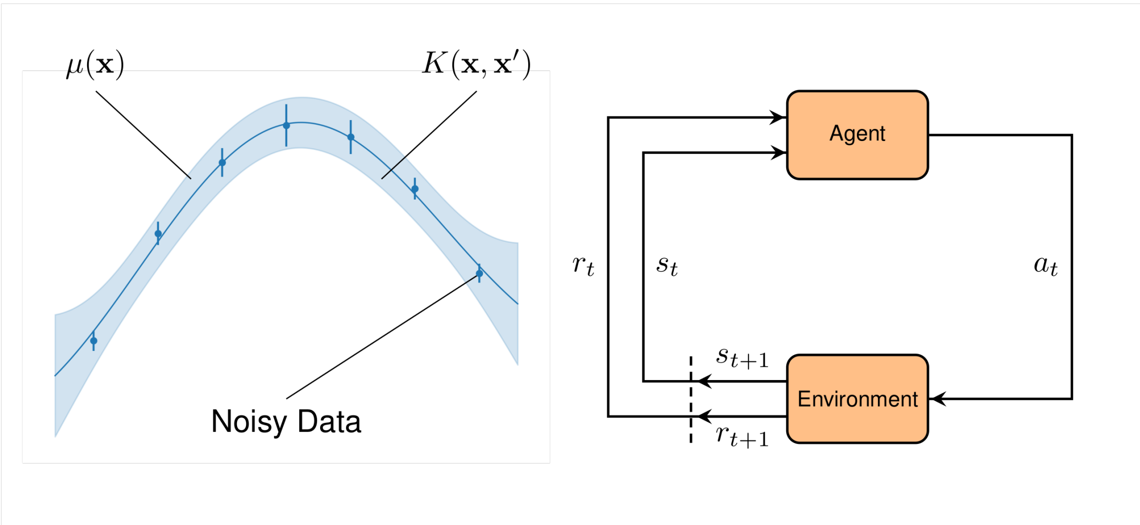

Figure 5: Bayesian Optimization and Reinforcement Learning. Bayesian Optimization approximates noisy data by fitting a Gaussian Process (left), which provides the mean function $\mu(\mathbf{x})$ and the kernel function $K(\mathbf{x}, \mathbf{x}')$ for covariance. Reinforcement Learning (right) involves an agent interacting with the environment, where the agent, in state s_t, optimizes an action a_t to receive a reward r_t.

Training ML models

I will train the ML models on the ray-tracing simulations of the XES experiment. To simulate the proposed XES setup, the project will utilize the expertise of Alessandro Scordo, responsible for the VOXES project, as his experience is particularly valuable because he has demonstrated that ray-tracing simulations produce reliable results for acquired spectra [17], which can be effectively utilized to simulate them. SHADOW-XOP with the OASYS environment [18] has been identified as the optimal choice for this task. This package, given the input lineshape under investigation at the source point, can perform ray-tracing through all elements in the setup and display the simulated spectrum on the detector. The input lineshape will be obtained using experimental core level widths data from the literature [19] or from ab-initio with Dirac-Fock codes [20] as I have previously calculated X-ray emission spectra from first-principles [21]. **The importance of performing ray-tracing simulations is twofold: firstly, it will serve as a preliminary test for the optimization procedure; secondly, it will be a core component of the framework used for training the deep learning models required for both BO and RL**. I will use a Convolutional Neural Network (CNN) to extract features from the acquired spectra and provide them as input to the BO loop, as well as to determine the policy for the RL algorithm. This will involve training the CNN on simulated spectra obtained through the ray-tracing simulations. Eventually, I will explore the possibility of training on historical data to bridge the gap between simulation and real data (see Figure 3). I will use `pytorch` to facilitate the interface with the BO component. The trained models will be validated using standard ML metrics, such as the F1-score and ROC curves. Here, my previous experience with ML techniques and databases, although applied in a different context, will be valuable and transferable [22–24].

4 Organization

Here I describe the timeline of the project, divided into **Work Packages** (WP) and listing the **Milestones** (M) and **Deliverables** (D):

WP1 - Developing the Optimization Framework (Months 1-12)

I will develop and test the optimization framework on available test datasets. Concurrently, together with A. Scordo, I will start ray-tracing simulations of the experimental setup. With the framework and the simulations, I will start the training of the ML models, to then apply the optimization on the simulations as the first preliminary test of the approach.

- **D1.1 Technical note on the optimization framework (Month 6):** Note on the implementation, development, and testing of the optimization framework.
- **D1.2 Documentation on raytracing simulations (Month 6):** Documentation on the accuracy of the raytracing simulations to simulate spectra for the XES setup.
- **D1.3 Summary of training and validation of ML models (Month 9):** Summary of the training of ML models performed on the ray-tracing simulations.
- **M1.1 Framework implemented (Month 6):** The optimization framework is fully implemented and operational.

WP2 - Assembling the experimental setup (Months 1-12)

The integration of the pixel detector and its motorized stages will be conducted here. Once the design is completed, I will assemble the setup with the help of M. Bazzi, and the components will be interfaced within the framework with the aid of A. Clozza.

- **D2.1 Report on the integration of detector (Month 6):** Detailed document on the integration process of the pixel detector into the setup plus the technical drawing of the support.
- **D2.2 Summary on measurements with the new detector (Month 12):** Summary of the initial measurements taken with the integrated pixel detector.

- **M2.1 Detector integrated (Month 6):** The pixel detector, motors, and design are successfully integrated into the setup.

WP3 - Optimization of XES measurement (Months 13-21)

With the participation of A. Scordo, I will begin with SO optimization, focusing on optimizing one quantity at a time, and then, MO methods will be applied to balance acquisition time, precision, and resolution at the same time. Finally, a benchmark of the two methods, BO/RL, will be performed, together with a scan of multiple emission lines in the 5-20 keV range for different elements (Ti, Zr, Fe...).

- **D3.1 Results of optimization with the new setup (Month 21):** Report of the results from the application of the optimization framework to the new setup.

WP4 - Validation and Applications (Months 19-24)

I will conduct XES measurements for the Kβ line of iron in a diluted liquid sample to validate the optimized setup, comparing the results with previous VOXES apparatus measurements [25]. Together with C. Curceanu, I will identify in-situ applications at LNF to extend the synergies of the project within a broader context. Finally, I will visit the AXIm group in London to present the idea and its potential.

- **D4.1 XES Validation Report (Month 21):** Report on XES measurements for Kβ iron in diluted liquid, validating the optimized setup against previous data.
- **D4.2 Final report on the OPTIMAL impact and synergies (Month 24):** Document detailing the final impact and synergies of the developed XES setup and the framework.
- **M4.1 In-situ facilities identified (Month 24):** An applicable use case for the framework has been recognized within the LNF facilities.

WP5 - Dissemination (Months 10-24)

OPTIMAL will result in the publication of at least two scientific papers in peer-reviewed journals. Here, I include the participation of all the personnel involved in international conferences and the organization of a workshop.

- **D5.1 Technical Article (Month 18):** A technical article describing the setup, the interface to the framework, and examples of optimization.
- **D5.2 Validation Article (Month 24):** An article showcasing the framework and its validation through the XES setup.

The FTE and Gantt Chart of the project are reported in the Table (1) and in Figure (6).

Name	Units	FTE	WP involved
Simone Manti	INFN, LNF	1.0	All
Alberto Clozza	INFN, LNF	0.3	WP2
Alessandro Scordo	INFN, LNF	0.3	WP1,WP3
Massimiliano Bazzi	INFN, LNF	0.1	WP2
Catalina Curceanu	INFN, LNF	0.1	WP4,WP5
Totale	-	**1.8**	-

Table 1: FTE of the project. Research units, members, and participation percentages in terms of Full Time Equivalent (FTE) of the OPTIMAL project.

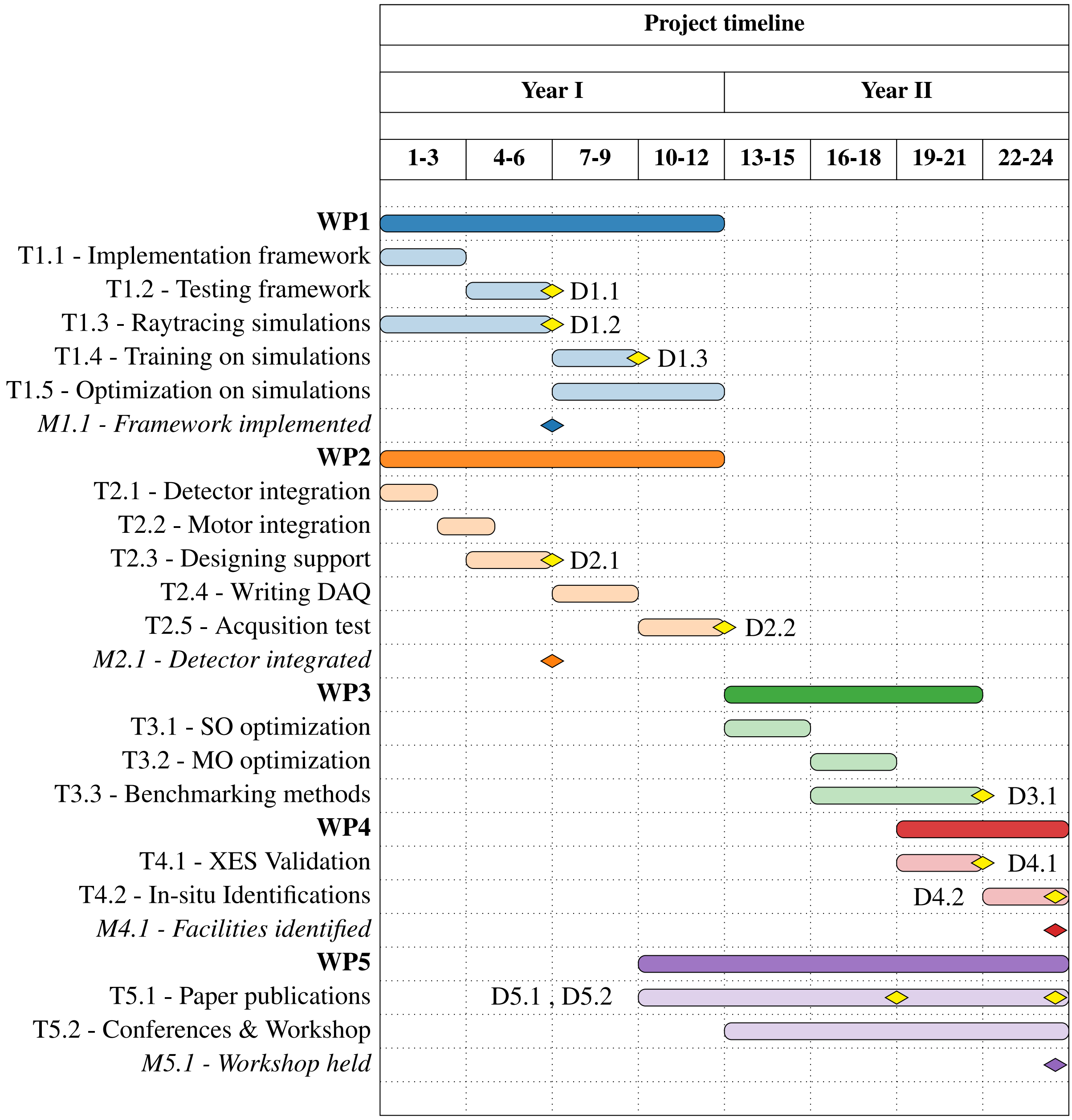

Figure 6: Gantt Chart of the project. Gantt Chart for the two years of the project, divided into WPs and relative tasks. All deliverables are indicated as yellow diamonds, while milestones have the same color as the associated WP.

5 Synergies with external projects

The OPTIMAL project will establish extensive synergies with various external projects, institutions and companies

INFN Scientific Commissions and Collaborations

- **VOXES:** The project will establish strong synergies with the VOXES project at LNF, as is providing substantial in-kind contributions and knowledge exchange regarding the development of the XES setup.
- **SPHINX:** Another project that will benefit from these synergies is SPHINX within CSN5. Although this project uses X-rays for holography rather than spectroscopy, it faces similar challenges in optimizing the alignment of optical components for imaging. Given the time-limited application of this project on the beamline, the need for rapid optimization tasks is even more critical, highlighting the importance of efficient alignment processes.
- **DAΦNE-Light and EuAPS:** Within LNF, synergies will be developed with DAΦNE-Light and EuAPS. DAΦNE-Light, utilizing a conventional X-ray source for X-ray Fluorescence (XRF), is well-suited for transitioning from a XES setup to a different application. Indeed, despite not using a crystal monochromator, XRF in energy-dispersive mode, demands precise tuning of the X-ray source intensity to avoid detector dead time during positional scanning for elemental analysis. The OPTIMAL project can optimize this by dynamically adjusting the intensity in real-time while scanning the sample. Similarly, EuAPS could benefit from the project through the optimization of both the pulsed laser's parameters and the alignment of optical components, thereby enhancing overall performance and efficiency.

National and International Research Institutions

- **AXIm (London):** Synergies will be established at the international level with Advanced X-ray Imaging (AXIm) group in London, where Silvia Cipiccia expressed vivid interest in the project [see Figure (7)] towards the application for fast X-ray ptychography.

Industry and Private Sector

The project's integration of optimization methods and machine learning has attracted significant interest from companies supplying components for the setup [see Figure (7)].

- **ADVACAM (Prague):** The project will collaborate with ADVACAM, the company that provides the pixel detector. The inclusion of a ML framework to process and learn from the acquired images has attracted considerable attention, as it has potential applications beyond spectroscopy, extending to X-ray imaging.
- **CRISEL (Rome):** The supplier of the STANDA motors, has shown interest in the project. Their interest lies in how the framework will control the motors to enhance optimization, highlighting the broader applicability of the project's innovations. Collaboration with Crisel will also contribute to the development of more precise and reliable motor control systems.

Synergies with these external projects and institutions will provide valuable resources, expertise, and validation opportunities, significantly enhancing the overall success and impact of the OPTIMAL project.

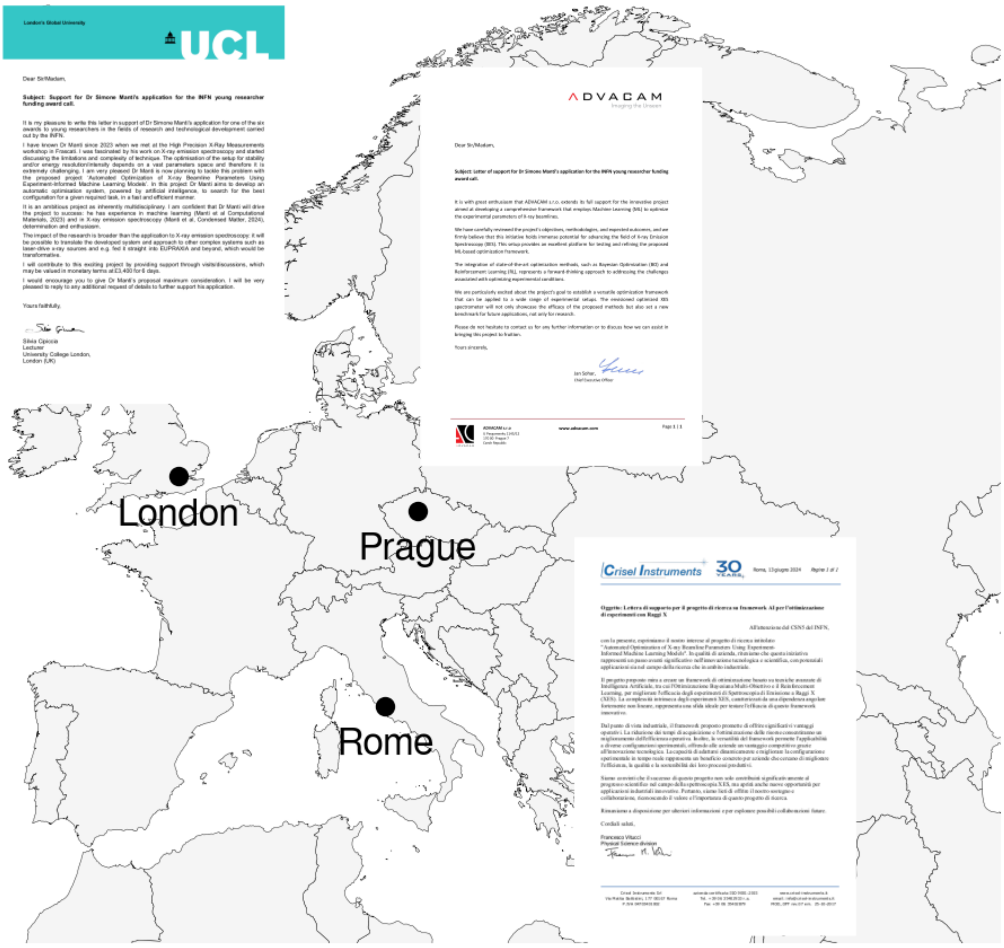

Figure 7: Support Letters for the OPTIMAL project. Letters of support for the OPTIMAL project. On (*top left*), the letter from Silvia Cipiccia from the AXIm research group based in London, on (*top right*) the letter from ADVACAM in Prague and on (*bottom right*) from Crisel in Rome.

6 Impact

The OPTIMAL project will have a significant impact on the INFN community and the X-ray scientific community at large, advancing the state of the art in optimizing experimental parameters of X-ray beamlines through the use of ML.

Impact on the INFN and scientific community as follows:

- **Framework:** Having a generalized framework applicable to X-ray beamlines will enhance internal collaboration across different experiments and significantly improve those that currently lack AI-guided optimization. Additionally, the framework will streamline data collection and organization, making it easier to share data among various research groups and improving the reproducibility of results. The project will, where possible, release software as open-source, contributing to the broader scientific community and facilitating further research. Finally, having a framework that employs state-of-the-art ML models for optimization will attract researchers and students, enhancing their skill sets and career prospects.
- **X-ray beamlines:** The impact of optimizing X-ray beamlines is significant as this project aims to enhance the technology currently in use. By automating and intelligently controlling the parameters, the project will reduce human errors and improve data quality and reproducibility. We anticipate a tenfold reduction in acquisition time for optimization, leading to substantial time and cost savings. This is particularly important for beamlines where the available time slots are limited and optimizing component alignment efficiently is crucial. Additionally, there are social benefits to consider. Reducing the time and human interaction required in these processes is advantageous, especially in environments where personnel are exposed to ionizing radiation. Improved working conditions and safety are essential outcomes of this project.
- **XES setup:** The proposed application to the XES setup exemplifies the significant impact of the project. It will clearly demonstrate the practical benefits of integrating advanced optimization techniques with existing experimental infrastructure. For instance, an optimized XES spectrometer will be highly beneficial in material science, which increasingly relies on high-throughput elemental analysis for numerous samples. This enhancement will streamline the analysis process, improving efficiency and accuracy in material characterization.

7 Costs

The cost and in-kind cost for each requested item are detailed in Table (2).

8 Risk assessment

Here I will outline the possible risks in the project with the possible mitigation.

1. **Inaccuracy of Ray-Tracing Simulations (T1.3)**
   - **Risk:** The ray-tracing simulations may not accurately describe the XES experiment, as they are typically devoted to X-ray beamlines optics and sources, but they are not optimized for simulating mosaic crystals. Such materials have been indeed included in the XOP package only by M. Sanchez del Rio [18].
   - **Mitigation:** Maintain close contact with the developer of the SHADOW-XOP program, Manuel Sanchez del Rio, to seek assistance to improve the accuracy of the simulations.
2. **Data Integration and Feature Extraction Issues (T1.4)**

Year	Activity	Cost in Euro	In-Kind Cost in Euro
Year I	XES setup	0	120,000
Year I	Pixel Detector*	48,000	0
Year I	Motors*	10,000	0
Year I	LNF support	1,000	0
Year I	Workstation*	8,000	0
Year I	Travels	5,000	0
Total Year I	-	**72,000**	**120,000**
Year II	Portable GPU*	3,000	0
Year II	Travel to AXIm	0	4,000
Year II	Target elements	2,000	0
Year II	Publications	3,000	0
Year II	Conferences	5,000	0
Year II	Travels	5,000	0
Year II	Workshop	10,000	0
Total Year II	-	**28,000**	**4,000**
Total Cost	-	**100,000**	**124,000**

Table 2: OPTIMAL Budget Summary. The table provides an overview of the project's budget over two years, detailing both direct costs and in-kind contributions for various activities and resources. For items marked with a * a quotation is included in the project.

- **Risk:** Integrating data from different sources and accurately extracting features from XES spectra using deep learning may present technical challenges.
- **Mitigation:** Use historical data to bridge gaps between simulated and real-world data.

3. **Delays in Ordering or Integrating New Equipment (T2.1, T2.2, T2.3)**

   - **Risk:** Potential delays in the procurement or integration of the pixel detector and other new equipment could affect project timelines.
   - **Mitigation:** Use a CCD from the VOXES project as a temporary measure to initiate measurements until the pixel detector integration is complete. This CCD, although less efficient and with too small sensitive area, will allow for continued progress on the project.

4. **Single/Multi-Objective Optimization Complexity (T3.1,T3.2)**

   - **Risk:** Balancing multiple objectives (e.g., acquisition time, energy precision, and resolution) in the optimization process may be complex and computationally intensive.
   - **Mitigation:** Regularly evaluate and adjust the optimization strategies based on intermediate results. Engage with experts in optimization and machine learning to refine and enhance the optimization algorithms. Specifically, contact Lucio Anderlini (Florence Section of INFN), whose expertise in artificial intelligence and deep learning can provide valuable insights and support for the project.

In conclusion, the successful completion of this project will create a strong and flexible optimization framework that can be applied to various applications, including DAΦNE-Light and EuAPS, significantly expanding the possibilities for X-ray beamline research.

References

[1] P. I. Frazier, "A tutorial on bayesian optimization," *arXiv preprint arXiv:1807.02811*, 2018.

[2] R. Garnett, *Bayesian optimization*. Cambridge University Press, 2023.

[3] P. Liu, *Bayesian Optimization: Theory and Practice Using Python*. Springer, 2023.

[4] B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, and N. De Freitas, "Taking the human out of the loop: A review of bayesian optimization," *Proceedings of the IEEE*, vol. 104, no. 1, pp. 148–175, 2015.

[5] R. Roussel, A. L. Edelen, T. Boltz, D. Kennedy, Z. Zhang, X. Huang, D. Ratner, A. S. Garcia, C. Xu, J. Kaiser, *et al.*, "Bayesian optimization algorithms for accelerator physics," *arXiv preprint arXiv:2312.05667*, 2023.

[6] R. Roussel, A. Hanuka, and A. Edelen, "Multiobjective bayesian optimization for online accelerator tuning," *Phys. Rev. Accel. Beams*, vol. 24, p. 062801, Jun 2021.

[7] E. Cisbani, A. Del Dotto, C. Fanelli, M. Williams, M. Alfred, F. Barbosa, L. Barion, V. Berdnikov, W. Brooks, T. Cao, *et al.*, "Ai-optimized detector design for the future electron-ion collider: the dual-radiator rich case," *Journal of Instrumentation*, vol. 15, no. 05, p. P05009, 2020.

[8] J. Duris, D. Kennedy, A. Hanuka, J. Shtalenkova, A. Edelen, P. Baxevanis, A. Egger, T. Cope, M. McIntire, S. Ermon, and D. Ratner, "Bayesian optimization of a free-electron laser," *Phys. Rev. Lett.*, vol. 124, p. 124801, Mar 2020.

[9] S. Jalas, M. Kirchen, P. Messner, P. Winkler, L. Hübner, J. Dirkwinkel, M. Schnepp, R. Lehe, and A. R. Maier, "Bayesian optimization of a laser-plasma accelerator," *Phys. Rev. Lett.*, vol. 126, p. 104801, Mar 2021.

[10] C. Benmore, T. Bicer, M. K. Chan, Z. Di, D. a. Gürsoy, I. Hwang, N. Kuklev, D. Lin, Z. Liu, I. Lobach, *et al.*, "Advancing ai/ml at the advanced photon source," *Synchrotron Radiation News*, vol. 35, no. 4, pp. 28–35, 2022.

[11] C. T. Cakir, C. Bogoclu, F. Emmerling, C. Streli, A. G. Buzanich, and M. Radtke, "Machine learning for efficient grazing-exit x-ray absorption near edge structure spectroscopy analysis: Bayesian optimization approach," *Machine Learning: Science and Technology*, vol. 5, no. 2, p. 025037, 2024.

[12] T. Morris, M. Rakitin, A. Islegen-Wojdyla, Y. Du, M. Fedurin, A. Giles, D. Leshchev, W. Li, P. Moeller, B. Nash, *et al.*, "A general bayesian algorithm for the autonomous alignment of beamlines," *arXiv preprint arXiv:2402.16716*, 2024.

[13] S. Kandel, L. Rebuffi, W. Cha, R. J. Harder, X. Shi, M. J. Highland, M. G. Frith, and M. J. Cherukara, "Bayesian optimization for autoalignment of an x-ray focusing system," in *Flat Optics: Components to Systems*, pp. JW2A–32, Optica Publishing Group, 2023.

[14] D. Marcato, L. Bellan, D. Bortolato, M. Comunian, A. Galatà, F. Gelain, V. Martinelli, G. Savarese, G. Susto, *et al.*, "Demonstration of beam emittance optimization using reinforcement learning," in *Proceedings of the 14th International Particle Accelerator Conference*, 2023.

[15] F. Hutter, H. Hoos, and K. Leyton-Brown, "An efficient approach for assessing hyperparameter importance," in *International conference on machine learning*, pp. 754–762, PMLR, 2014.

[16] M. Balandat, B. Karrer, D. R. Jiang, S. Daulton, B. Letham, A. G. Wilson, and E. Bakshy, "BoTorch: A Framework for Efficient Monte-Carlo Bayesian Optimization," in *Advances in Neural Information Processing Systems 33*, 2020.

[17] V. De Leo, A. Scordo, C. Curceanu, M. Miliucci, and F. Sirghi, “Reflection efficiency and spectra resolutions ray-tracing simulations for the voxes hapg crystal based von hamos spectrometer,” *Condensed Matter*, vol. 7, no. 1, 2022.

[18] L. Rebuffi and M. S. del Rio, “Oasys (orange synchrotron suite): an open-source graphical environment for x-ray virtual experiments,” in *Advances in Computational Methods for X-Ray Optics IV*, vol. 10388, pp. 169–177, SPIE, 2017.

[19] J. Campbell and T. Papp, “Widths of the atomic k–n7 levels,” *Atomic Data and Nuclear Data Tables*, vol. 77, no. 1, pp. 1–56, 2001.

[20] J. Desclaux, “A multiconfiguration relativistic dirac-fock program,” *Computer Physics Communications*, vol. 9, no. 1, pp. 31–45, 1975.

[21] K. Piscicchia, S. Donadi, S. Manti, A. Bassi, M. Derakhshani, L. Diósi, and C. Curceanu, “X-ray emission from atomic systems can distinguish between prevailing dynamical wave-function collapse models,” *Phys. Rev. Lett.*, vol. 132, p. 250203, Jun 2024.

[22] S. Manti, M. K. Svendsen, N. R. Knøsgaard, P. M. Lyngby, and K. S. Thygesen, “Exploring and machine learning structural instabilities in 2d materials,” *npj Computational Materials*, vol. 9, p. 33, Mar 2023.

[23] M. N. Gjerding, A. Taghizadeh, A. Rasmussen, S. Ali, F. Bertoldo, T. Deilmann, N. R. Knøsgaard, M. Kruse, A. H. Larsen, S. Manti, T. G. Pedersen, U. Petralanda, T. Skovhus, M. K. Svendsen, J. J. Mortensen, T. Olsen, and K. S. Thygesen, “Recent progress of the computational 2d materials database (c2db),” *2D Materials*, vol. 8, p. 044002, jul 2021.

[24] F. Bertoldo, S. Ali, S. Manti, and K. S. Thygesen, “Quantum point defects in 2d materials - the qpod database,” *npj Computational Materials*, vol. 8, p. 56, Apr 2022.

[25] S. Manti, M. Miliucci, A. Scordo, R. Bedogni, A. Clozza, M. Iliescu, G. Moskal, K. Piscicchia, A. Porcelli, D. Sirghi, F. Sirghi, and C. Curceanu, “Detecting iron oxidation states in liquids with the voxes bragg spectrometer,” 2023.